■ PathFinder

Personalized Career Recommendation System Using Machine Learning

Comprehensive Technical & Architectural Report (1st Report)

Project Title	Personalized Career Recommendation System Using Machine Learning
Platform Name	PathFinder (Classes 7 to 12)
Champion Model	V9.5-Champion XGBoost Classifier (with ColumnTransformer Preprocessor)
Hit@1 Ranking Quality	96.03% (Top 1 Accuracy)
Hit@3 / Hit@5 Quality	99.64% / 99.89% (Top 3 & Top 5 Recall)
Mean Reciprocal Rank (MRR)	0.9781
Normalized DCG (NDCG@5)	0.9211
Classification Accuracy / ROC-AUC	81.07% / 0.8537 ROC-AUC
Catalogue & Question Pool	1,203 Unique Careers (33 Domains) | 413 Questions (19 Sections)
Database Backend	MySQL 8.x (18 Relational Tables) / SQLAlchemy ORM
Automated Test Suite	83 / 83 Tests Passing (100% Test Coverage)
Document Date	30 August 2026

1. Executive Summary & Problem Statement

In the Indian secondary school landscape (Classes 7 through 12), career selection is frequently constrained by limited exposure, societal biases, and reliance on generic, unscientific assessments. Students rarely receive guidance tailored to both their intrinsic cognitive competencies and their evolving disciplinary interests. **PathFinder** was engineered to resolve this challenge through an intelligent, personalized, and psychometrically validated platform powered by supervised machine learning.

PathFinder incorporates an end-to-end architecture consisting of an adaptive psychometric questionnaire (50–55 questions selected from 413 across 19 sections), a 22-dimensional cognitive/interest scoring service, an 11-feature engineering vector pipeline, and a champion **XGBoost Classifier**. The system evaluates every student against all 1,203 careers in its structured catalogue, filters candidates against domain prerequisite thresholds, and outputs ranked recommendations with actionable skill roadmaps.

96.03% Hit@1 Precision	**99.64%** Hit@3 Coverage	**99.89%** Hit@5 Coverage
0.9781 MRR Score	**0.9211** NDCG@5 Score	**0.8537** ROC-AUC Score
81.07% Accuracy	**1,203 Careers** Career Pool	**83/83 Passing** Unit Tests

2. Full-Stack System Architecture & Database Schema

PathFinder employs a modular Flask application factory architecture with five dedicated Blueprints and an isolated database layer built on **MySQL 8.x** via SQLAlchemy ORM. The data model comprises 18 normalized relational tables initialized through a unified setup.sql script:

Table Name	Domain Responsibility	Key Columns & Foreign Keys
users	Authentication & Security	id, username, email, password_hash (bcrypt), is_admin
students	Student Profile Registry	id, user_id (FK), student_code, class_level (7-12), stream, board
academic_scores	17-Subject Mark Records	student_id (FK), math, science, physics, chemistry, bio, cs, english... overall_pct
assessment_sessions	Assessment State Machine	id, student_id (FK), status (in_progress/completed), current_q, completion_pct
student_answers	Question Response Auditing	id, assessment_id (FK), question_id (FK), selected_option, numeric_value, time_tak
assessment_scores	22-Dimension Normalized Scores	assessment_id (FK), mathematical_ability, logical_reasoning... social_interest
questions	Master Question Bank (413 Qs)	id, section_id (FK), text, question_type (MCQ/RATING/SCENARIO), skill_category,
question_sections	Assessment Section Taxonomy	id (1-19), name, description, display_order
question_options	Scored Answer Choices (1805)	id, question_id (FK), option_text, option_value, score (0-100)
career_domains	33 High-Level Domains	id, domain_name, description, icon, display_order, is_active
career_subdomains	Domain Specialization Tracks	id, domain_id (FK), name, description
career_clusters	Occupational Sub-clusters	id, subdomain_id (FK), name, description
careers	1,203 Career Catalogue	id, career_name, domain_id (FK), cluster_id (FK), min_education, work_environmen
career_skills	Skill Competency Matrix	career_id (FK), skill_name, proficiency_level, is_core

Table Name	Domain Responsibility	Key Columns & Foreign Keys
career_subjects	Academic Prerequisites	career_id (FK), subject_name, importance_level
career_education	5-Stage Academic Roadmaps	career_id (FK), stage_order, title, description, typical_degrees
career_recommendations	Persisted ML Predictions	assessment_id (FK), career_id (FK), rank_position, compatibility_score, probability

3. Adaptive Assessment & Selection Engine

The AssessmentSelectionService implements dynamic, grade-appropriate, and attempt-differentiated question selection. Rather than presenting a static questionnaire, the engine queries the 413-question master pool and constructs a balanced assessment matching the student's cohort quotas:

Grade Cohort	Eligible Classes	Question Target	Difficulty Strategy	Stream Customization
Middle School	Classes 7 & 8	50 Questions	Easy (60%) + Medium (40%)	General Foundation Track
Secondary	Classes 9 & 10	52 Questions	Easy (30%) + Med (50%) + Hard (20%)	Stream Exploration Track
Higher Secondary	Classes 11 & 12	55 Questions	Med (50%) + Hard (35%) + Easy (15%)	PCM / PCB / Commerce / Humanities

Section Quotas (19 Sections): Guaranteed minimum question delivery across every assessment ensures complete cognitive coverage:

Sec #	Assessment Section	Min (7-8)	Min (9-10)	Min (11-12)	Core Target Competency
1	Academic Profile	2	2	2	Study orientation & learning style
2	Mathematical Ability	5	5	6	Quantitative computation, algebra & geometry
3	Logical Reasoning	4	4	5	Pattern recognition, deductive logic & series
4	Scientific Thinking	4	4	5	Hypothesis isolation & causal deduction
5	Problem Solving	3	3	3	Constraint resolution & optimization
6	Analytical Thinking	2	3	3	Data interpretation, tables & charts
7	Communication	2	2	2	Verbal clarity & comprehension
8	Creativity	2	2	2	Divergent ideation & aesthetic intuition
9	Digital Ability	2	3	3	Computational logic & technology literacy
10	Learning Ability	2	2	2	Cognitive agility & conceptual adaptation
11	Spatial Ability	2	2	2	3D mental rotation & structural visualization
12	Practical Ability	2	2	2	Hands-on mechanical & tool dexterity
13	Core Interests	8	8	10	Tech, Medical, Business, Arts, Research, Social
14	Activities & Hobbies	4	4	4	Extracurricular behavioral engagement
15-19	Teamwork, Leadership & Work Prefs	6	6	6	Work environment, teamwork & autonomy

4. 22-Dimension Psychometric Scoring Pipeline

Upon test submission, ScoringService computes normalized percentage scores (0.0 to 100.0) across 22 independent cognitive, interest, and behavioral dimensions. The calculation normalizes earned option points against total possible points per skill category, applies SKILL_ALIASES mappings, and dynamically infers unmeasured dimensions via correlation baselines:

Category	Dimensions Tracked	Calculation Methodology & Score Bands
8 Primary Abilities	Math, Logical, Scientific, Problem Solving, Analytical, Communication, Creativity, Digital	MCQ/Scenario option score / max category score × 100. Normalized to [0, 100].
7 Extended Traits	Learning Ability, Memory, Observation, Spatial, Practical, Teamwork, Leadership	Rating (1-5) normalized to (0-100) and multi-select fractional scoring.
7 Core Interests	Technology, Science, Healthcare, Business, Creative, Research, Social	Direct questionnaire scoring + Correlated baseline inference (e.g. Research = 0.6×
Guidance Bands	5 Educational Guidance Bands	Excellent (80.5-100) | Good (60.5-80.4) | Average (40.5-60.4) | Low (20.5-40.4) | V

5. Feature Engineering & Machine Learning Pipeline

The FeatureBuilder transforms student assessment scores and the 1,203-career catalogue into a high-dimensional feature contract. For every candidate career, a 19-dimensional feature vector is generated using vectorized NumPy operations:

#	Feature Column	Mathematical Formulation	Role in Inference
1	ability_match_component	mean(100 - |student_ability - required_ability|) across 8 pairs	Primary cognitive fit metric
2	interest_match_component	Weighted mean(100 - |student_int - required_int|) with 1.5× top-3 boost	Disciplinary-interest alignment
3	academic_match_component	Student overall academic % across 17 subjects	Academic rigor compatibility
4	learning_match_component	Normalized learning_ability score (0-100)	Growth & adaptability factor
5	composite_alignment_index	0.45×Ability + 0.35×Interest + 0.10×Academic + 0.10×Learning	Unified holistic compatibility index
6	ability_interest_synergy	(ability_match × interest_match) / 100.0	Non-linear synergy interaction
7	ability_interest_gap	|ability_match - interest_match|	Identifies potential misalignment
8	min_core_match	min(ability_match, interest_match)	Bottleneck constraint detector
9	max_core_match	max(ability_match, interest_match)	Peak capability indicator
10	harmonic_core_match	2 × (A × I) / (A + I + 1e-5)	Harmonic balance penalty metric
11	geometric_core_synergy	sqrt(max(0, A × I))	Geometric mean alignment
12	holistic_synergy	(A × I × Academic × Learning) ^ 0.25	Full-spectrum 4-factor synergy
13-14	age, class	Student age (10-30) and Class level (7-12)	Demographic scaling features
15-19	career_name, domain, sub, clus, stream	OrdinalEncoded categorical taxonomy & stream alignment	Domain-specific prior learning

Domain Prerequisite & Compliance Filtering:

To guarantee real-world educational feasibility, CareerRecommendationEngine enforces domain threshold constraints configured in config.yaml. For example, Healthcare careers require minimum scientific (60%) and mathematical (60%) thresholds. Candidate careers passing compliance checks are sorted with strict priority: threshold_pass DESC → probability DESC → ability_match DESC → interest_match DESC.

6. Model Performance & Empirical Evaluation

The champion model was rigorously validated against held-out multi-cohort evaluation datasets. The table below summarizes both ranking quality and binary classification metrics:

Evaluation Metric	Score	Benchmark Target	Practical System Impact
Hit@1 (Top-1 Accuracy)	96.03%	≥ 90.0%	Primary recommended career is directly compatible in 96% of cases.
Hit@3 (Top-3 Recall)	99.64%	≥ 95.0%	Near-perfect likelihood of target career in top recommendations.

Evaluation Metric	Score	Benchmark Target	Practical System Impact
Hit@5 (Top-5 Recall)	99.89%	≥ 98.0%	Virtually eliminates catastrophic recommendation failure.
Mean Reciprocal Rank (MRR)	0.9781	≥ 0.90	Exceptional rank-ordering of relevant career candidates.
NDCG@5 (Discounted Gain)	0.9211	≥ 0.85	High rank quality with optimal top-heavy positioning.
Classification Accuracy	81.07%	≥ 80.0%	Balanced binary decision accuracy across candidate pairs.
Precision (Weighted)	83.72%	≥ 80.0%	High confidence in positive compatibility predictions.
Recall (Weighted)	91.66%	≥ 85.0%	Minimizes false negatives; compatible options are captured.
F1-Score	87.51%	≥ 82.0%	Optimal balance between precision and recall.
ROC-AUC / PR-AUC	0.8537 / 0.9349	≥ 0.80	Superior discriminative capability in imbalanced candidate spaces.

7. REST API Endpoints & Automated Test Suite

PathFinder exposes a structured REST API conforming to standard JSON envelope contracts. The entire backend is backed by an automated **83-test test suite** running under isolated in-memory SQLite environments:

API Endpoint	HTTP Verb	Auth Guard	Functionality
/api/health	GET	Public	Exposes health status for DB, model, preprocessor, and catalogue.
/api/model/info	GET	Public	Authoritative model version metadata, features, and ranking metrics.
/api/questions/<class_level>	GET	Public	Fetches adaptive question set matching student grade and stream.
/api/assessment/start	POST	Student	Initializes assessment session and persists selected question IDs.
/api/assessment/answer	POST	Student	Auto-saves individual student answer with millisecond audit timestamp.
/api/assessment/submit	POST	Student	Finalizes test, triggers ScoringService and XGBoost recommendations.
/api/recommendations/<id>	GET	Student/Admin	Retrieves persisted Top-K career recommendations with explanations.
/api/student/profile	GET/PUT	Student	Fetches and updates student demographic details and 17 subject marks.

Test Suite Breakdown (83 / 83 Tests Passing — 100% Success):

Test Module	Test Count	Key Verification Scope
test_ml_model_loading & prediction	7 Tests	Singleton integrity, ColumnTransformer, thread-safety, probabilities.
test_ml_feature_builder & recommendation	5 Tests	19-feature vector assertions, catalogue loading, Top-K ranking.
test_ml_concurrency, performance & security	8 Tests	Multi-threaded stress test, SHA-256 integrity, path traversal safety.
test_assessment_workflow & selection	16 Tests	End-to-end session lifecycles (Classes 7-12), attempt differentiation.
test_scoring & profile synthesis	10 Tests	Deterministic 0/50/80/100 scoring, alias mapping, baseline inference.
test_auth, admin, user history & career import	24 Tests	Bcrypt security, role guards, student audits, CSV ingestion.
test_career explorer & details	13 Tests	Search filtering, 33 domains, 5-stage educational roadmaps.

8. Conclusion & Future Roadmap

The PathFinder platform demonstrates the efficacy of combining rigorous psychometric assessment design with modern gradient-boosted decision trees (XGBoost) for educational guidance. By evaluating 1,203 careers simultaneously across 19 engineered features and enforcing domain prerequisite thresholds, PathFinder achieves an outstanding **96.03% Hit@1** precision and **0.9781 MRR** while ensuring zero identity leakage (enforced student_id exclusions).

Future enhancements will focus on integrating real-time entrance examination eligibility mappings (JEE, NEET, CUET, CLAT), scholarship discovery APIs, and LLM-powered conversational counseling agents to complement quantitative career recommendations.

PathFinder — Helping Every Student Find Their Best Path Forward.

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